

FDK™ Toolkit, Education Domain Fairness Audit

External Validation Report: LSAC Bar Passage Prediction Dataset

Post-Hoc No-Code UX Hardening and Real-Dataset Validation

Dataset: LSAC National Longitudinal Bar Passage Study (Wightman, 1998), bar_pass_prediction.csv extract, 22,407 individuals

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Abstract

This report documents the validation of the FDK™ Toolkit's Education fairness-audit pipeline against a real, publicly available benchmark dataset, the LSAC National Longitudinal Bar Passage Study. Education is one of two FDK domains (with Justice) that predate the toolkit's now-standard no-code dropdown UX pattern and required post-hoc hardening rather than fresh construction. This review found and fixed a duplicate form-field name defect and completed client-side statistical filtering for the upload page's target-column dropdown, previously unfiltered and header-only, a known gap shared across all seven FDK domains. A mid-review edit-application error briefly broke file upload entirely and was diagnosed and corrected. Real-dataset validation then surfaced a semantically wrong auto-detected prediction column (the same category of defect, bug class #11, first identified during the Justice domain's COMPAS validation), resolved by deriving a genuine prediction column from the dataset's own existing model-probability output rather than training a new model. An initial derivation threshold produced a degenerate, near-constant column and was corrected using a base-rate-matched threshold. The group column's keyword-based detection also selected a coarser numeric race encoding over the dataset's standard categorical one, resolved via manual override, and a naive substring keyword match caused an unrelated enrollment-status column to be misdetected as a timestamp. With all defects resolved, the Education pipeline produced a complete, independently verified 26-metric audit on the race1 protected attribute, revealing a substantial real disparity in bar-passage prediction between Black and White law students.

1. Background and Purpose

Education, along with Justice, predates the no-code dropdown UX pattern established across Finance, Health, Hiring, Business, and Governance, and required retrofitting rather than fresh construction. This review continued the project's established file-by-file methodology from a prior session's handoff SOP document, reviewing upload_education.html, fdk_education.py, fdk_education_pipeline.py, and auto_confirm_education.html in that order, one file at a time.

The LSAC Bar Passage Prediction dataset was selected for this validation: a real, extensively studied dataset in the algorithmic-fairness literature, predicting bar-examination passage from law-school and demographic records. Its use here serves the same purpose as every other domain's real-dataset work in this project: confirming the FDK™ Toolkit's own correctness against real, messy data rather than synthetic test fixtures.

2. Dataset Description

Property	Value
Total records	22,407 individuals; 16 rows with missing race1 excluded during audit (sample size 22,391)
Source	LSAC National Longitudinal Bar Passage Study (Wightman, 1998); bar_pass_prediction.csv, a dataset widely used in the algorithmic fairness literature
Group column used	race1 (5 categories: white, black, hisp, asian, other)
Ground-truth column	pass_bar (bar examination passage, binary)
Prediction column	predicted_pass (derived from dnn_bar_pass_prediction via a base-rate-matched threshold, not a newly trained model)
Probability column	dnn_bar_pass_prediction (the dataset's own existing model-probability output, range 0.331–0.9997)
Race1 group sizes	White: 18,716 · Black: 1,343 · Hisp: 1,027 · Asian: 897 · Other: 408 (16 rows missing excluded)

One property of this dataset is directly relevant to Finding 6 below: the dataset also contains a separate, more granular numeric race column (race, 8 numeric codes), which the keyword-based detection layer matches equally well and, being earlier in the file's own column order, is selected ahead of race1 whenever no manual override is applied.

3. Findings From Real-Dataset Validation

Finding 1, Duplicate form-field name risked silently dropping the upload page's manual override

upload_education.html's hidden target_column field and its manual-override <select> both submitted under the same form field name, target_column. Because the hidden field appeared first in the form and was kept in sync via JavaScript on every interaction path, the defect was currently masked rather than actively broken, but any future code path updating one without the other would silently drop the user's actual selection, the same pattern documented as bug class #7 across the FDK Toolkit's other domains.

Remediation: renamed the hidden field to target_column_fallback, matching the convention already used correctly in the Business domain (select owns the submitted name; hidden field is a fallback only), making the select the single source of truth.

Finding 2, Upload-page target dropdown listed every raw column, unfiltered

The upload page's JavaScript read only the file's header row via FileReader and populated the target-column dropdown with every raw column name, with no cardinality or binary-ness filtering, a known, previously acknowledged gap shared across all seven FDK domains, not unique to Education. A reference file initially offered as an "already-fixed" example from the Business domain was checked directly and found to have the identical, unaddressed limitation.

Remediation: built client-side CSV row-sampling (up to 1,000 data rows per file, quote-aware parsing) and a strict {0,1}-binary column classifier mirroring the server's own is_binary_column convention, filtering the dropdown to plausible target columns only, with a graceful fallback to the full column list (with an on-screen warning) if no binary column is found in the sample. The same client-side candidate set also filters whatever the server's /api/detect-columns endpoint itself returns, so the dropdown stays consistent regardless of what that endpoint returns.

Verification: verified in an isolated jsdom (headless DOM) simulation against a dataset shaped like the real one, correctly narrowing an 8-column file down to its one true binary column, and correctly falling back to the unfiltered list (with a warning) on a file with no binary column at all.

Finding 3, A mid-review edit-application error broke file upload entirely

After Findings 1 and 2's fixes were applied, the upload page could no longer accept a CSV file at all. Diagnosis traced this to a stray copy of Finding 1's replacement HTML having been pasted inside the `<script>` block itself, producing a JavaScript syntax error that silently prevented the entire script, including the file-input event listener, from ever attaching.

Remediation: removed the single misplaced line; the surrounding, correctly applied fixes were otherwise untouched.

Verification: re-parsed the corrected script with Node's syntax checker to confirm the file was free of errors before redeployment.

Finding 4, Auto-detected prediction column was statistically valid but semantically wrong

The first real-dataset audit run auto-selected `parttime` (a student's part-time enrollment flag) as the prediction column, because the dataset's genuine model-prediction column, `predicted_pass`, did not natively exist; it had not yet been derived, and the statistical fallback layer defaulted to the first available binary column it encountered in file order. This is the same category of defect documented as bug class #11 in the project's SOP: a column can be statistically valid for a role while being semantically meaningless, first identified during the Justice domain's COMPAS validation.

Remediation: derived a genuine prediction column, `predicted_pass`, from the dataset's own existing model-probability output (`dnn_bar_pass_prediction`), rather than training a new model from scratch.

Finding 5, Initial derivation threshold produced a degenerate, near-constant prediction column

The first derivation attempt thresholded `dnn_bar_pass_prediction` at a fixed 0.5 cutoff. Because that probability column's floor value (0.331) was already well above 0.5, this produced a prediction column with only 1 negative value out of 22,407 rows, functionally constant, and incapable of supporting meaningful disparity metrics.

Remediation: recalculated the threshold to match the true base rate of `pass_bar` (94.78%), producing a non-degenerate binary column (95.18% predicted-positive rate; confusion matrix of 165 true negatives, 1,004 false positives, 914 false negatives, and 20,324 true positives against the actual outcome).

Verification: confirmed the corrected CSV's `predicted_pass` distribution and confusion matrix directly before it was used in any audit run.

Finding 6, Group column detection selected a coarser numeric encoding over the dataset's standard categorical one

The dataset contains two race-related columns, `race` (8 numeric codes) and `race1` (the standard 5-category white/black/hisp/asian/other encoding used in this dataset's fairness literature). Both satisfy the keyword-matching detection layer's group criteria equally; `race` was selected only because it appears earlier in the file's own column order, not because of any semantic preference.

Remediation: applied via the confirmation page's manual override to `race1`; a purely code-level fix was not pursued, since column order is not a reliable general signal for domain preference across arbitrary datasets.

Finding 7, Naive substring keyword-matching misdetected an unrelated column as a timestamp

The timestamp-detection keyword list matches keywords as raw substrings of column names; the generic keyword `time` matched inside `fulltime` (a student full-time-enrollment flag), which was consequently misdetected as a timestamp column. This went undetected because pandas' `pd.to_datetime` does not raise on a 0/1 integer column; it silently interprets the values as 1970-epoch nanosecond timestamps.

Remediation: replaced the substring check with token-aware matching (splitting column names on non-alphanumeric separators and requiring a whole-token match), and added a second, independent guard rejecting any parsed timestamp column whose values collapse entirely into 1970-ish dates.

Verification: confirmed no column in the real dataset is misdetected as a timestamp after the fix, and specifically confirmed fulltime and parttime are excluded. Re-ran the real audit and confirmed timestamp: null with every core disparity metric numerically unchanged to the last decimal place, isolating the fix's effect to only the metrics that depend on feature-set composition.

Verification of fixes

Check	Before fix	After fix
Upload-page target dropdown	Every raw column listed, unfiltered	Filtered to sampled binary columns, with safe fallback
Hidden/select field name collision	Shared name (target_column) on both	Distinct names; select is sole source of truth
File upload functionality	Broken (JS syntax error)	Restored; confirmed via Node syntax check
Prediction column used in first real audit	parttime (semantically meaningless)	predicted_pass (derived from the dataset's own model output)
predicted_pass distribution (0.5 threshold)	22,406 positive / 1 negative (degenerate)	21,328 positive / 1,079 negative (base-rate matched)
Group column used	race (8 numeric codes)	race1 (standard 5-category encoding)
timestamp column detected	fulltime (false positive)	null (correctly, no genuine timestamp column exists)
Core disparity metrics, pre/post timestamp fix	statistical_parity_difference 0.14380419507194342	statistical_parity_difference 0.14380419507194342 (unchanged)

4. Fairness Findings on the LSAC Bar Passage Prediction Dataset

With all seven defects resolved, the Education pipeline produced one complete, independently verified 26-metric audit on the race1 protected attribute, on the full cleaned population (n=22,391).

Cut 1: race1

Metric	Value	Interpretation
disparate_impact_ratio	0.851	Comfortably clears the conventional 0.80 four-fifths threshold
statistical_parity_difference	0.144	A 14.4 percentage-point selection-rate gap between the highest and lowest group
equal_opportunity_difference	0.120	A 12.0 percentage-point true-positive-rate gap among qualified students
selection_rates_by_group (Black)	0.819	Lowest predicted-pass rate of the five groups analyzed
selection_rates_by_group (White)	0.963	Highest predicted-pass rate
composite_fairness_score	0.367	Toolkit's own classification: POOR, despite disparate_impact_ratio clearing the four-fifths rule, the other disparity metrics do not

This finding illustrates a point worth stating plainly: disparate_impact_ratio clearing the conventional four-fifths threshold is not, on its own, evidence of fairness, statistical_parity_difference and equal_opportunity_difference both show a real, substantial gap between Black and White students on this dataset and this model, and the toolkit's own composite score correctly reflects that as POOR rather than deferring to the single metric that looks acceptable.

5. Scope Limitations of This Validation Run

- This validation used a single protected-attribute cut (race1); a second cut (gender/sex, both present in this dataset) was not run in this session, unlike the two-cut convention used in every other domain's validation to date.
- predicted_pass was derived from the dataset's own existing dnn_bar_pass_prediction model output via a base-rate-matched threshold, rather than training a new model from scratch; these findings describe that existing model's behavior on this LSAC dataset, not a claim about current bar-admission practices generally.
- auto_confirm_education.html was reviewed against the same bug classes checked in the upload page (duplicate field names, hardcoded URLs, display: none ancestor traps, dead hidden inputs) and found clean, but was not exercised end-to-end with a live audit run in the same way the upload page and pipeline were.
- fairness_drift_index (temporal stability) could not be computed for this dataset, since it contains no genuine timestamp column; its reported value of 1.0 is confirmed to be the pipeline's deliberate neutral default in the absence of a timestamp, not a measured result.
- This is a well-established academic dataset (Wightman, 1998) rather than a live production system; findings describe the dataset and its own existing prediction model, not any real current bar-admission decision process.

6. Conclusion

The Education domain fairness-audit pipeline's no-code user experience has been brought up to the same standard as the toolkit's five later-developed domains, and validated against a real, independently collected, extensively cited benchmark dataset. Seven genuine defects were identified, root-caused, fixed, and verified, including one, the semantically wrong prediction column, that is the same category of defect first identified during the Justice domain's own COMPAS validation, and one, the mid-review edit-application error, that was caught and corrected before it reached production. The pipeline's resulting fairness findings show a substantial, real disparity in bar-passage prediction between Black and White law students on this dataset, with the toolkit's own composite score correctly classifying the result as POOR even where a single headline metric (disparate_impact_ratio) alone would have suggested otherwise.

Appendix, Reference Materials

Dataset: bar_pass_prediction.csv (LSAC National Longitudinal Bar Passage Study, Wightman, 1998)

Modified source files: upload_education.html (duplicate-name fix; client-side row-sampling statistical filter; syntax-error correction); fdk_education.py (token-aware timestamp keyword matching; epoch-year guard)

Validation reports referenced: education_audit_report_20260723_211552.json (parttime mis-detected as prediction, pre-fix); education_audit_report_20260723_212925.json (corrected column mapping, pre-timestamp-fix); education_audit_report_20260723_214015.json (fully resolved)

Related reports: Governance_API_External_Validation_Report.docx (source of this report's format); Justice_COMPAS_External_Validation_Report.docx; Finance_API_External_Validation_Report.docx; Health_API_External_Validation_Report.docx; Hiring_API_External_Validation_Report.docx; Business_API_External_Validation_Report.docx